

Algebra 1
Unit 2
Lesson 9 Homework

Name Answer Key
Date _____
Period _____

1. Use the area model to find $(-4.5x^4y^3 + 2.4x^3y^2 - 3.6x^2y) \div (3xy)$ where $x \neq 0$ and $y \neq 0$.

$$-1.5x^3y^2 + 0.8x^2y - 1.2x$$

	$-1.5x^3y^2$	$0.8x^2y$	$-1.2x$
$3xy$	$-4.5x^4y^3$	$2.4x^3y^2$	$-3.6x^2y$

Find each quotient.

2. $(\frac{3}{4}x^3 + \frac{1}{2}x^2) \div (-\frac{1}{2}x^2)$ where $x \neq 0$.

$$-1.5x - 1$$

3. $(4a^3 + 8a^2 - 20a) \div (4a)$ where $a \neq 0$

$$a^2 + 2a - 5$$

4. $(-10k^6r^4 + 6k^2r^4 - 42k^3r^3) \div (2k^2r)$ where $k \neq 0$ and $r \neq 0$

$$-5k^4r^3 + 3r^3 - 21kr^2$$

5. $(-17w^6 + 13w^3 - 18w^2) \div (2w)$ where $w \neq 0$

$$-8.5w^5 + 6.5w^2 - 9w$$

6. $\frac{12x^4 - 30x^3 + 6x^2 - 18}{6x}$ where $x \neq 0$

$$2x^3 - 5x^2 + x - \frac{3}{x}$$

7. $\frac{14p^4+42p^3+21p^2-7p}{7p}$ where $p \neq 0$

$$2p^3 + 6p^2 + 3p - 1$$

8. $\frac{2t^4-2t^3+12t^2-7t}{2t}$ where $t \neq 0$

$$t^3 - t^2 + 6t - \frac{7}{2}$$

9. $\frac{2.8x^3+4.8+3.2x^2-0.8}{0.4x}$ where $x \neq 0$

$$7x^2 + \frac{10}{x} + 8x$$

10. $\frac{15y^3+45y^2+35y^2}{-5y^2}$ where $y \neq 0$

$$-3y - 16$$